

Alina Du

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EDUCATION

Columbia University

Expected May 2028

B.S. Computer Science, Financial Engineering and Math Minor | **GPA: 3.9/4.0**

New York, NY

- Coursework: Software Quality and Design, Data Structures, Machine Learning, Optimization
- Activities: Teaching Assistant, SWE, WiCS, Columbia Lioness Quantitative, Rewriting the Code

SKILLS & AWARDS

Languages: Python, Java, TypeScript, SQL, HTML

ML/Data: PyTorch, Scikit-learn, Hugging Face, NumPy, Pandas, SciPy

Frameworks: React/React Native, Next.js, FastAPI, LangChain, LangGraph

Backend: AWS, GCP, Vercel, Supabase, Firebase, PostgreSQL, MongoDB, Git

Awards: Jane Street WiSE (top 60); MIT Math Prize for Girls; 2x AIME Qualifier; 2x Hackathon Winner; Waterloo National Math Scholarship (15 recipients); 8x Waterloo math contest top 1%

EXPERIENCE

Machine Learning Research Intern

May 2026 - Present

Columbia University

New York, NY

- Presented *Research Poster* at Jane Street and Columbia Research Fair; co-authored paper in progress
- Improved reliability from 0.918 to 0.965 across 50 held-out trees via an adaptive risk-thresholding policy that allocates a fixed verification budget to the highest-risk reasoning steps
- Benchmarked 7 LLMs on 5K+ claims and found systematic overconfidence on errors, motivating structural verification

Research Assistant – Data Engineering

Sep 2026 - Present

Columbia University

New York, NY

- Co-develop the lab's econ-LLM experiment platform, building the study UI and prompt conditions
- Build Python ETL pipelines to clean, normalize, and analyze experimental research data

Software Engineer Intern

Mar 2026 - Jun 2026

Intelligence Cubed

New York, NY

- Designed and implemented an invite-code access-control system gating the deployment flow, enforced on both frontend and backend to prevent unauthorized provisioning
- Served Cloudflare-hosted Kimi models via an Anthropic-compatible API across 5+ services
- Built one API for users to deploy AI models to AWS, GCP, and Cloudflare from a single backend

Software Engineer Intern

Aug 2025 - Mar 2026

MeloMed – Health Tech Startup

New York, NY

- Built React Native interfaces delivering sentiment-driven recommendations to 1,200+ users
- Reduced infrastructure costs by 23% through serverless AWS deployment and caching optimizations
- Improved query latency through PostgreSQL indexing and database performance tuning

PROJECTS

Pac-Manhattan | JavaScript, Three.js, Python, Shapely, Gemini API, ElevenLabs

- Built a 3D NYC game with 171K buildings in Three.js and A* pathfinding on a 27K-edge graph
- Built a Shapely pipeline in Python that turns OpenStreetMap data and aerial imagery into per-borough JSON maps
- Built an LLM content pipeline with Gemini to check riddles for leaked answers, ElevenLabs narration

Hack the Heart | Python/FastAPI, Next.js, OpenAI/Anthropic API, MongoDB, pytest

- Ran rule-based NLP (VADER, hedging) on full chat histories; LLM reads the newest 10K messages
- Stored date turns in MongoDB by (date ID, turn index) so failed dates resume without duplicates
- Imported chats from multiple sources with message-ID dedup, keeping only the user's messages